Recommendations — osteosarcoma-mets-dll3-h7r2

Ranked therapeutic options with likelihood, toxicity, mechanism risk, and key references

Generated 2026-05-13

Recommendations — osteosarcoma-mets-dll3-h7r2

Profile snapshot

- **Primary site:** bone
- **Histology:** osteosarcoma
- **Stage:** IV
- **ECOG:** 1
- **Age band:** 18-29
- **Sex:** unknown
- **Biomarkers:** DLL3 RNA expression elevated; protein-level membrane status unknown (rna_only); PRAME RNA expression elevated; protein-level expression and HLA presentation status unknown (rna_only)
- **Efficacy/toxicity weight:** 0.85

4 ranked therapeutic options across 2 targetable features. Biomarker workup steps live in the standalone Target validation paths report.

DLL3-targeting interventions

1. tarlatamab via NCT06788938 (UCCC-01 basket) *(conditional on dll3_ihc positive)*

Likelihood of effect: Cross-tumor extrapolation: SCLC OS HR 0.60 (DeLLphi-304); ORR ~40% (DeLLphi-301). Osteosarcoma efficacy is the open question NCT06788938 will answer.

Toxicity burden: Moderate (CRS ~50% mostly G1-2; CRS G≥3 ~1%; ICANS-like ~10%; inpatient cycle-1 step-up dosing required)

Counter-productive MoA:Moderate (On-mechanism CNS bystander T-cell activation drives ICANS; possible DLL3 antigen-loss escape on repeated dosing)

Overall:The only DLL3-directed option when IHC is positive — preference-aligned but cross-tumor translation untested; foreclosed if IHC negative.

Key references: NCT06788938 · PMID 37861218 · PMID 40454646

2. SHR-4849 / IDE849 via NCT07174583 (IDEAYA pan-tumor DLL3 basket) *(conditional on dll3_ihc positive)*

Likelihood of effect: Speculative. No published clinical data for SHR-4849. TOP1-inhibitor-payload ADC precedent (T-DXd, sacituzumab govitecan) suggests the payload class can produce response signals, but DLL3-specific efficacy is the open question.

Toxicity burden: Unknown — no published clinical data. TOP1-inhibitor ADC class effects expected: cytopenias, GI toxicity, possible ILD/pneumonitis (DXd-class ADCs carry an ILD signal worth monitoring).

Counter-productive MoA:Moderate (ADC bystander toxicity to DLL3-low normal tissue; antigen-loss escape on repeated dosing; PBD-payload class shadow (Rova-T TAHOE) does not directly apply but informs the toxicity-budget framing.)

Overall:Mechanism-distinct second DLL3 pathway pending sponsor confirmation of osteosarcoma eligibility — relevant if the tarlatamab path is foreclosed or fails.

Key references: NCT07174583

Pipeline context — not currently enrollable

Other agents in this target class drawn from the case dossier. The patient cannot currently enroll in these trials, but they are included for context on the broader modality landscape.

Intervention	Modality	Phase	Enrolling indication	Recruitment	Trial
alveltamig	trispecific	3	relapsed SCLC	recruiting	NCT07189455
zocilurtatug pelitecan	ADC	3	small-cell lung cancer	recruiting	NCT07218146
obrixtamig	BiTE	3	extensive-stage SCLC (1L)	recruiting	NCT07472517
gocatamig	trispecific	1/2	DLL3-expressing SCLC, large-cell NEC, neuroendocrine prostate cancer, GEP-NEC	active not recruiting	NCT04471727
peluntamig	bispecific_other	1/2	SCLC, LCNEC, NEPC, GEP-NEC, EP-NEC	recruiting	NCT05652686
QLS31904	BiTE	1	advanced solid tumors with DLL3 expression	unknown	NCT05461287
DLL3-CAR-NK cells	CAR-NK	1	extensive-stage SCLC	unknown	NCT05507593
clesitamig	trispecific	1	SCLC, neuroendocrine carcinoma	completed	NCT05619744
LB2102	CAR-T	1	extensive-stage SCLC, LCNEC of lung	active not recruiting	NCT05680922
FZ-AD005	ADC	1	advanced solid tumors, SCLC, LCNEC	recruiting	NCT06424665
IBI3009	ADC	1	small-cell lung cancer	recruiting	NCT06613009
^{22}Ac-ABD147	radioligand	1	SCLC, large-cell neuroendocrine carcinoma	active not recruiting	NCT06736418
^{1}Lu-DTPA-SC16.56		1	neuroendocrine tumors / carcinomas (lung, prostate)	not yet recruiting	NCT06941480
^{22}Ac-ETN029	radioligand	1	SCLC, LCNEC, NEPC, GEP-NEC	recruiting	NCT07006727
rovalpituzumab tesirine (discontinued)	ADC	3	small-cell lung cancer (2L)	completed	NCT03061812
SC-002 (discontinued)	ADC	1	small-cell lung cancer	terminated	NCT02500914

AMG 119 (discontinued)	CAR-T	1	small-cell lung cancer	unknown	NCT03392064

Evidence in detail

tarlatamab

Ahn et al. *N Engl J Med* 2023

- **Disease context:previously treated SCLC** (2L+) — SCLC after platinum-based therapy and ≥1 prior line; ECOG 0-1; brain mets allowed if treated
- **n:** 220
- **Toxicities:CRS** (grade any): 51%; **CRS** (grade >=3): 1%; **ICANS-like neurologic event** (grade any): 10%; **treatment-related AE** (grade >=3): 30%; **treatment discontinuation due to AE** (grade any): 3%
- **Efficacy:ORR** 40% (95% CI 29-52); **DoR** not reached months; **PFS** 4.9 months; **TRAE_rate** 30%

Mountzios et al. *N Engl J Med* 2025

- **Disease context:SCLC after platinum-based chemo** (2L) — Platinum-resistant or platinum-sensitive SCLC; ECOG 0-1
- **n:** 509
- **Toxicities:CRS** (grade any): 56% (n=254); **CRS** (grade >=3): 1% (n=254); **ICANS-like neurologic event** (grade any): 12% (n=254); **treatment-related AE** (grade >=3): 24% (n=254); **neutropenia** (grade >=3): 3% (n=254)
- **Efficacy:HR_OS** 0.6 HR (95% CI 0.47-0.77); **OS** 13.6 months; **OS** 8.3 months; **PFS** 4.2 months; **PFS** 3.2 months; **TRAE_rate** 24%

PRAME-targeting interventions

1. IMA203 (PRAME-TCR-T) via NCT03686124 (Immatics ACTengine pan-solid basket) *(conditional on prame_ihc_hla positive)*

Likelihood of effect: Strong. ORR 54% (95% CI 34-73) in pan-solid PRAME+ HLA-A*02:01+ cohort (Wermke 2024 Lancet Oncol, n=28). Sarcoma cohorts on protocol; osteosarcoma fit unconfirmed but eligibility is mechanism-driven, not tumor-restricted.

Toxicity burden: High (CRS ~100% — G3-4 ~10%; ICANS-like ~25%; uniform post-Cy/Flu cytopenias; one treatment-related death in published cohort; CAR-T-style infusion infrastructure required)

Counter-productive MoA:Moderate (On-target / off-tumor toxicity to PRAME-expressing normal testis is class-managed (testis is immune-privileged); CRS / neurotoxicity from T-cell activation is the main mechanism-level risk)

Overall:Best-evidenced PRAME pathway with pan-solid sarcoma-inclusive basket; conditional on PRAME IHC + HLA-A*02:01 typing both confirming.

Key references: NCT03686124 · PMID 38821093

2. IMC-P115C via NCT07156136 (Immunocore next-gen PRAME ImmTAC pan-tumor) *(conditional on prame_ihc_hla positive)*

Likelihood of effect: Speculative. No published clinical data for IMC-P115C; relies on brenetafusp class precedent (ORR ~9% in heavily pretreated melanoma; durable in subset).

Toxicity burden: Moderate (ImmTAC class: CRS ~85% mostly G1-2; rash ~70%; transient hypotension; pre-medication-managed)

Counter-productive MoA:Moderate (ImmTAC on-target / off-tumor signal in PRAME-expressing normal tissue (low; testis-restricted); CRS from T-cell activation)

Overall:Mechanism-class alternative to IMA203 in the PRAME space, contingent on PRAME + HLA confirmation and sponsor confirmation osteosarcoma is in scope.

Key references: NCT07156136 · PMID 39007852

Pipeline context — not currently enrollable

Other agents in this target class drawn from the case dossier. The patient cannot currently enroll in these trials, but they are included for context on the broader modality landscape.

Intervention	Modality	Phase	Enrolling indication	Recruitment	Trial
brenetafusp (IMC-F106C) (basket/biomarker)	bispecific_other	1/2	advanced solid tumors expressing PRAME (cutaneous and uveal melanoma, ovarian, lung, sarcoma cohorts)	active not recruiting	NCT04262466
IMA402 (PRAME-TCER) (basket/biomarker)	bispecific_other	1/2	refractory / recurrent solid tumors expressing PRAME	recruiting	NCT05958121
brenetafusp (IMC-F106C)	bispecific_other	3	previously untreated HLA-A*02:01-positive advanced cutaneous melanoma	recruiting	NCT06112314
BPX-701 (discontinued)	CAR-T	1/2	AML, MDS, uveal melanoma	terminated	NCT02743611

Evidence in detail

IMA203 (PRAME-TCR-T)

Wermke et al. *Lancet Oncol* 2024

- **Disease context:PRAME-positive HLA-A*02:01-positive refractory solid tumors (basket: melanoma, sarcomas including synovial sarcoma, ovarian, NSCLC, head and neck, uterine, others)** (2L+) — Heavily pretreated PRAME+ HLA-A*02:01+ patients; subset of synovial sarcoma cohort enrolled; lymphodepleting Cy/Flu prior to infusion
- **n:** 28
- **Toxicities:CRS** (grade any): 100% (n=28); **CRS** (grade >=3): 11% (n=28); **ICANS-like neurotoxicity** (grade

any): 25% (n=28); **post-Cy/Flu cytopenia** (grade >=3): 100% (n=28); **treatment-related death** (grade 5): 4% (n=28)
- **Efficacy:ORR** 54% (95% CI 34-73); **DoR** not reached months; **PFS** 5.1 months; **TRAE_rate**

IMC-P115C

Hamid et al. *Cancer Discov* 2024

- **Disease context:PRAME-positive HLA-A*02:01-positive advanced cutaneous melanoma** (2L+) — Heavily pretreated melanoma; subset with prior immunotherapy; HLA-A*02:01-positive; PRAME-positive by IHC
- **n:** 132
- **Toxicities:CRS** (grade any): 85% (n=132); **CRS** (grade >=3): 3% (n=132); **rash** (grade any): 70% (n=132); **transient hypotension** (grade any): 35% (n=132); **any treatment-related AE** (grade >=3): 30% (n=132)
- **Efficacy:ORR** 9% (95% CI 4-17); **PFS** 2.1 months; **DoR** >6 months; **TRAE_rate** 30%